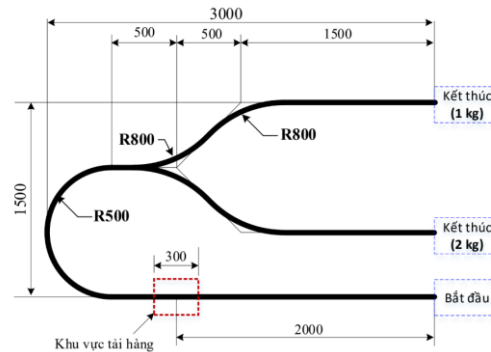


Exercise 1

1. Working Space for mobile robot is in following Figure (2 marks)

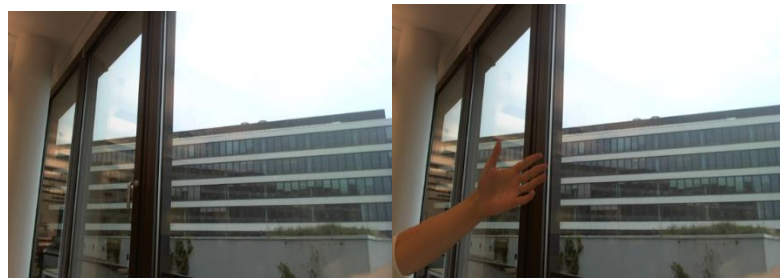


Feature needing to watch on robot has diameter 1mm. Chose appropriate Basler Camera and working distance. Assume that the camera is setup above and perpendicular to the working space. Remember to check the result on the Basler website.

2. A picture has the matrix as following with STT is the index of the group (2 marks)

STTx10	20	30	STTx10
80	STTx10	100	110
120	160	STTx10	150
220	230	240	250

- Equalize Histogram this picture in range 0-255, using 256 bin
 - Program using opencv lib to equalize histogram this picture and display histogram before and after equalize
 - threshold this picture using Otsu method with random initial guessing, Check again using opencv lib.
3. Two different pictures are shown in following figures (2 marks)



- Calculate Euclidean distance between two (R,G,B) of pictures (remember to scale the pictures to the same size)

$$\sqrt{(R_1 - R_2)^2 + (G_1 - G_2)^2 + (B_1 - B_2)^2}$$

b) Set threshold of Euclidean distance to see the hand.

4. Different pictures are shown in following figures (2 marks)



a) Program to store the number of white pixel on the circle with different Radius (1,5,10,15,20,25,30,35,40) and center of the circle is at the center of the picture to the TXT file (remember to scale the pictures to the same size)

b) Compare to identify the chessman

5. Extract color of lemon and determine the good, bad and acceptable lemon (2 marks)

